

# NeuroGAN: Synthetic Data Augmentation for Alzheimer's MRI Classification

## 1. Problem Framing

Alzheimer's disease is a progressive neurological disorder where early and accurate diagnosis is critical. Deep learning models trained on MRI scans have shown strong potential, but real-world medical datasets suffer from severe class imbalance. Certain disease stages are well represented, while others, often clinically important, contain very few labeled samples.

In the dataset used for this project, one disease category contained only 49 MRI samples, compared to hundreds or thousands in other classes. When trained on such imbalanced data, standard convolutional neural networks tend to favor majority classes, leading to misleadingly high accuracy while failing on underrepresented patient groups. Collecting additional medical data is expensive, slow, and ethically sensitive due to privacy constraints. This motivates the need for alternative approaches that improve model fairness without relying on new patient data.

## 2. Methods and Evaluation

**Baseline Classification:** A ResNet-18 model was trained on the original imbalanced dataset using 128 by 128 grayscale MRI slices. The data was split into training, validation, and test sets using an 80, 10, and 10 percent split. Model performance was evaluated using accuracy, macro F1 score, and per-class F1 score. Although the baseline model achieved strong overall accuracy, it showed weak and inconsistent performance on the minority class, confirming that data imbalance was the primary limiting factor rather than model capacity.

**Synthetic Data Generation:** To address this issue, a generative adversarial network named NeuroGAN was developed. NeuroGAN follows a DCGAN-style architecture and was trained exclusively on minority class MRI images. The generator learned to produce realistic 64 by 64 MRI scans from a 100-dimensional latent noise vector. Training was performed for 100 epochs using the Adam optimizer. Generated samples were visually inspected to ensure anatomical plausibility and diversity, and to avoid obvious artifacts or mode collapse.

**Augmented Training:** Synthetic minority class images were combined with the original dataset to create a more balanced training distribution. The same ResNet-18 architecture was retrained without any structural changes, ensuring a fair comparison between baseline and augmented models. Evaluation was performed on an unchanged held-out test set.

### **3. Results**

GAN-based augmentation led to a clear and measurable improvement in model performance. After training on the augmented dataset, the classifier achieved a test accuracy of 95.26 percent and a macro F1 score of 95.28 percent. Most notably, the minority class achieved a perfect F1 score of 1.00, indicating both perfect precision and recall. Confusion matrix analysis showed zero misclassifications for the minority class, demonstrating that the model successfully learned features that were previously underrepresented during training.

Qualitative inspection of synthetic images showed realistic structure and texture patterns consistent with real MRI scans. Remaining classification errors were primarily observed between other disease stages that share overlapping anatomical characteristics.

### **4. Limitations**

Despite strong performance gains, several limitations remain. The synthetic images were generated at a resolution of 64 by 64 pixels, which limits fine-grained anatomical detail. Synthetic data may also introduce subtle artifacts that are not immediately visible but could influence model behavior. Additionally, evaluation was conducted using a single dataset, without external validation across different scanners, institutions, or patient demographics. This system is intended strictly for research and must not be used for clinical diagnosis or decision making.

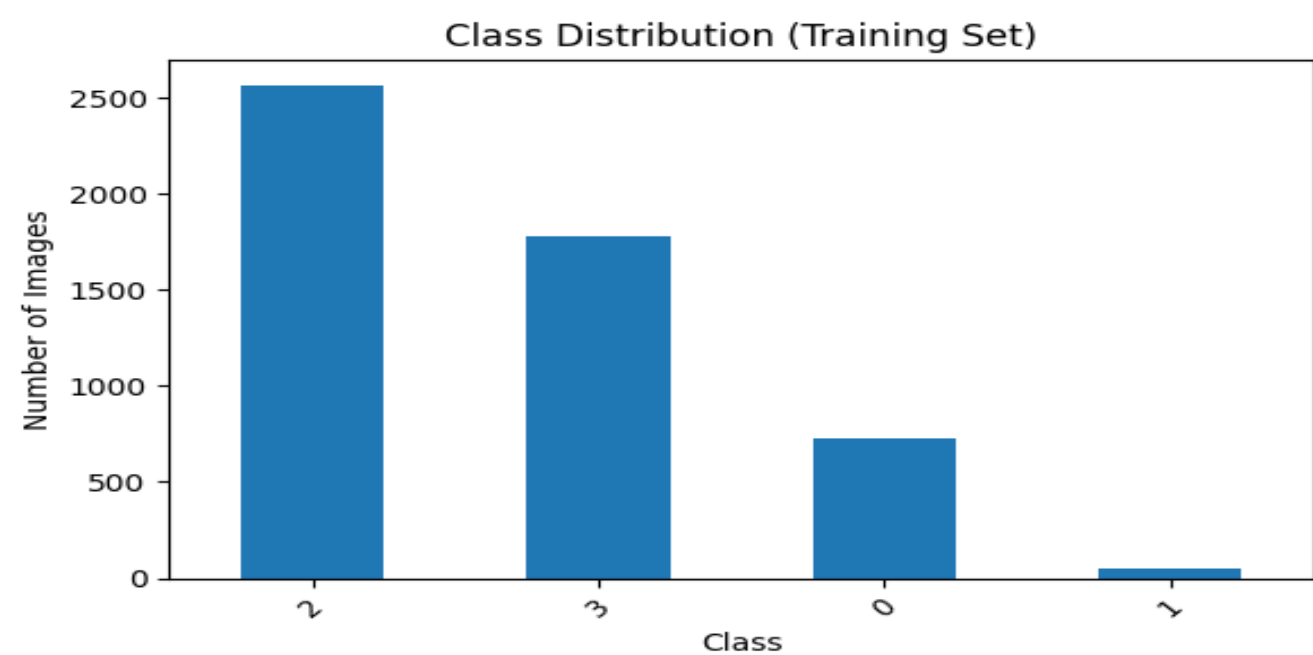
### **5. Next Steps and Societal Impact**

Future work will focus on improving the fidelity and generalization of synthetic data generation. Planned extensions include higher resolution conditional GANs, targeted augmentation of additional challenging classes, and the use of explainability techniques such as Grad-CAM to better understand model decisions. External validation on independent datasets will be essential to assess real-world robustness.

From a societal perspective, NeuroGAN highlights how synthetic data can reduce dependence on sensitive medical records, mitigate bias against underrepresented patient groups, and enable more equitable medical AI research. However, responsible use requires transparency, careful evaluation, and clear communication of limitations.

# Additional Results Visualizations:

## Dataset Distribution Before Augmentation



## Model Output and Metrics After Augmentation

